

# AI Network Security Analysis Report

## Corporate\_Data\_Breach

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### Executive Summary

The AI-powered network security monitoring system analyzed a Corporate\_Data\_Breach scenario involving 463 network packets over a 0.1 minute period. The system detected 0 security threats with an overall accuracy of 0.0%.

### Attack Narrative

#### *Phase 1: Initial Reconnaissance*

**Duration:** 15.0 minutes

**Packets Analyzed:** 300

**Threat Level:** low

Attackers perform comprehensive network scanning to identify vulnerable services

**AI Detection:** Normal activity detected during Initial Reconnaissance. Continuing monitoring.

**Recommended Actions:** Continue monitoring

#### *Phase 2: Spear Phishing Campaign*

**Duration:** 20.0 minutes

**Packets Analyzed:** 6

**Threat Level:** low

Targeted phishing emails sent to finance department with malicious attachments

**AI Detection:** Normal activity detected during Spear Phishing Campaign. Continuing monitoring.

**Recommended Actions:** Continue monitoring

#### *Phase 3: Lateral Movement*

**Duration:** 30.0 minutes

**Packets Analyzed:** 6

**Threat Level:** low

Compromised workstation used to access internal systems using stolen credentials

**AI Detection:** Normal activity detected during Lateral Movement. Continuing monitoring.

**Recommended Actions:** Continue monitoring

### *Phase 4: Data Staging*

**Duration:** 25.0 minutes

**Packets Analyzed:** 51

**Threat Level:** low

Sensitive files collected and compressed in preparation for exfiltration

**AI Detection:** Normal activity detected during Data Staging. Continuing monitoring.

**Recommended Actions:** Continue monitoring

### *Phase 5: Data Exfiltration*

**Duration:** 30.0 minutes

**Packets Analyzed:** 100

**Threat Level:** low

Large volumes of data transmitted to external command & control server

**AI Detection:** Normal activity detected during Data Exfiltration. Continuing monitoring.

**Recommended Actions:** Continue monitoring

## Machine Learning Insights

- ■ ML detected anomaly (score: 0.0)
- ■ Network health optimal (96%)
- ■ Top talker: external\_scanner (2 packets)
- ■ Top talker: external\_scanner (11 packets)
- ■ Top talker: external\_scanner (15 packets)
- ■ Top talker: external\_scanner (18 packets)
- ■ Top talker: external\_scanner (20 packets)
- ■ Top talker: external\_scanner (21 packets)
- ■ Top talker: external\_scanner (24 packets)
- ■ Top talker: external\_scanner (25 packets)

## Critical Findings

- 1. **Most Affected Devices:**
- 2. **Primary Attack Vectors:** none
- 3. **Data Exfiltration Risk:** Low
- 4. **Network Integrity Status:** Partially Compromised

## Response Effectiveness

| Metric                     | Value        |
|----------------------------|--------------|
| Mean Time to Detect (MTTD) | 0.00 seconds |
| Detection Rate             | 0.0%         |
| False Positive Rate        | 0.0%         |
| Overall Accuracy           | 0.0%         |

## Security Recommendations

- 1. Enhance detection capabilities with additional ML training
- 2. Conduct security awareness training for all users
- 3. Regular penetration testing and vulnerability assessments
- 4. Enable multi-factor authentication across all systems

## 5. Implement Zero Trust security model